Lesion-Anchored Prototypes: Reducing Artifact Shortcuts in Interpretable Melanoma Classification

Yousef Azizi Movahed
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
azizi.movahed@ut.ac.ir

Fatemeh Ziaeetabar*
Dept. of Computer Science, School of Mathematics, Statistics, and Computer Science, College of Science
University of Tehran, Tehran, Iran
fziaeetabar@ut.ac.ir

***Abstract*—Melanoma classifiers pick up artifacts such as dark corners, rulers and ink, and in a prototype network the explanation itself ("this part looks like that prototype") can point at the artifact. On ISIC 2019, dark corners and bright structures are associated with melanoma even after stratifying by source (Mantel–Haenszel odds ratios 1.555 and 1.751). Existing remedies supervise accuracy or activations; in dermoscopy, none controls *which* training patches a prototype may become. We propose lesion-anchored prototype learning with two components: a lesion-attraction objective on the normalized spatial attention of each prototype, and a projection step that admits only lesion cells without detected artifacts, so every prototype comes from the lesion by construction. Masks need no new annotation (Dice 0.94 on held-out expert masks, 0.92 on PH2), and inference none. With identical backbone, leakage-audited split and budget, our main model raises the test AUC of ProtoPNet from 0.825 to 0.865 (0.859 without its exploratory artifact-suppression term, a non-significant difference), not significantly different from the strongest mask-supervised ProtoPNet (0.867), and places all prototypes on the lesion, which no mask-loss weight or formulation achieves (9–18 of 20 on average). When artifacts are inserted around bit-identical lesion pixels, it roughly halves the decision-flip rate of ProtoPNet for dark corners (10.9%→5.2%) and rulers, flips fewer decisions under dark corners and ink than every mask-loss variant, and keeps its explanations the most stable (Spearman 0.92 vs. 0.63–0.88). The original mask loss generalizes better to external data, so the two approaches are complementary. Code and logs: https://github.com/YousefAMovahed/lesion-anchored-prototypes.**

***Index Terms*—explainable AI, prototypical part networks, shortcut learning, dermoscopy artifacts, melanoma, counterfactual evaluation**

I. Introduction

Convolutional networks can classify skin lesions as accurately as dermatologists [1], but they also learn *shortcuts* [2], features that correlate with the label in the training data without being related to the disease. In dermoscopy, surgical skin markings raise the melanoma probability of a deep classifier [3], models perform far above chance when the lesion is removed [4], [5], and dark corners, rulers, gel bubbles and ink are frequent and unevenly distributed across acquisition sites [6]–[8].

For clinical use, Rudin [9] argues for models that are interpretable by design. The prototypical part network (ProtoPNet) [10] compares image patches with learned prototypes, each projected onto a real training patch, so that a decision reads "this part looks like that part". Nothing, however, stops a prototype from being a piece of a ruler or a black corner; the shortcut then survives and is even presented as a reasonable explanation [11], and such prototypes have been observed in dermoscopy [12], [13]. **Research gap.** Prior work leaves three things open (Table I). (i) Debiasing methods for dermoscopy [7], [14], [15] improve robustness to artifacts, but their models are black boxes. (ii) For prototype networks, mask supervision penalizes activation outside annotated regions [12], [16], and human feedback removes or switches off unwanted prototypes [13], [17]; none of these methods constrains the *push* step, which decides which training patch a prototype finally *is*, and feedback acts only after a user has found the problem. (iii) Artifact insertion has tested the *decisions* of classifiers [13], [15], [18], but whether the *explanations* of a prototype network stay on the lesion when artifacts are added or removed has, to our knowledge, not been tested. **Contributions.** We fill these gaps as follows.

- **Lesion attraction on the normalized prototype attention.** An objective on the spatial attention of every prototype pulls it towards lesion tissue without shrinking the activations. It needs no new annotation and drives the accuracy gain: removing it lowers the test AUC from 0.865 to 0.830 ($p < 0.001$), and the main model gains 4.0 AUC points over ProtoPNet (3.4 with the two core components alone), not significantly different from the strongest mask-supervised ProtoPNet (the formulation of the closest prior work [12]), whose explanations rely three times more on artifacts.
- **Lesion- and artifact-constrained projection.** The push step admits only latent cells whose input patch is mostly lesion and free of detected artifacts, so every prototype comes from such a cell *by construction*, a guarantee that loss-based supervision does not give: no tested weight or formulation of the mask loss puts all 20 prototypes on the lesion in any run (on average 9–18). This step does not change the accuracy measurably (95% CI of the AUC difference $[-0.006, 0.005]$) but halves the share of the explanation captured by inserted dark corners

*Corresponding author: Fatemeh Ziaeetabar (fziaeetabar@ut.ac.ir).

and rulers. Foreground-restricted projection was proposed concurrently for natural images [19]; new here are the artifact exclusion, the combination with lesion attraction and the test against shortcuts.

- **A counterfactual test of decisions *and* explanations.** We insert four artifact types and remove real hair and ruler marks while every lesion pixel stays bit-identical, with an inpainting-only null control and a complementary test on unedited images, and measure how far both the decision and the prototype explanation move.
- **A leakage-controlled, mask-audited evaluation.** We group near-duplicates in addition to lesion identifiers, validate every mask source and the artifact detectors, compare eight configurations and four further variants of the mask-loss baseline with the same backbone over three seeds, use lesion-grouped statistics, and test on three external sets.

An artifact-suppression loss that we also explored brought no significant gain; we report it as a largely negative result, not as a contribution.

II. Related Work

Artifacts and shortcuts in dermoscopy. Bissoto *et al.* showed that ISIC models perform well when the lesion is occluded [4], annotated seven artifact types and found that simple debiasing does not remove the bias [5], and built artifact-based domain-generalization benchmarks [6]. Skin markings raise the melanoma score of a commercial-grade classifier [3], removing dark corners changes accuracy only marginally but moves activations onto the lesion [8], and pasted colored patches make a CNN call malignant lesions benign [18]. EPVT [7] learns domain prompts for artifact groups, Bevan and Atapour-Abarghouei [14] unlearn artifact information adversarially, and Reveal-to-Revise (R2R) [15] finds band-aids, rulers and skin markers semi-automatically, corrects the model with RRR, CDEP or ClArC and tests it on images with pasted artifacts. All of these models are black boxes.

Prototype-based interpretability. ProtoPNet [10] classifies by similarity to class-specific prototypes projected onto real training patches, which makes its reasoning inspectable [9]; the suggested similarity can, however, be misleading [11], which motivates testing explanations rather than taking them at face value [20].

Explanation supervision and prototype debugging. RRR [21] penalizes input gradients inside annotated irrelevant regions, and CDEP [22] contextual-decomposition importance, demonstrated on ISIC images with colored patches. IAIA-BL [16] adds a fine-annotation loss that penalizes prototype activation outside regions marked by radiologists. Closest to us, Correia *et al.* [12] apply this loss to melanoma-versus-nevus classification with lesion masks from a segmentation network; our PPN-FA baselines follow this approach, in its original formulation and in a weaker one (Section VI-G). ProtoPDebug [17] and the medical study of PIP-Net [13] let a user mark confounded prototypes, which are then forgotten or switched off. None of these methods constrains the projection step, and none tests prototype explanations under artifact interventions. Concurrently, Lin and Wright [19] restricted the projection to patches with sufficient foreground overlap on natural images; we add artifact exclusion and lesion attraction and test their effect on shortcut artifacts by ablation and counterfactuals.

TABLE I

Positioning against related work. Annot.: human input required; Interp.: interpretable by design; Where: constrains where the model looks; Prov.: constrains which training patches prototypes become; CF: artifact interventions on test images used to test decisions (D) and explanations (E). Lin and Wright [19] study natural images.

Method	Annot.	Interp.	Where	Prov.	CF
EPVT [7], Bevan [14]	image labels	–	–	–	–
R2R [15]	semi-auto.	–	✓	–	D, E
RRR [21], CDEP [22]	pixel masks	–	✓	–	–
ProtoPNet [10]	none	✓	–	–	–
IAIA-BL [16]	expert masks	✓	✓	–	–
Correia *et al.* [12]	auto. masks	✓	✓	–	–
ProtoPDebug [17]	user	✓	✓	post hoc	–
PIP-Net (med.) [13]	user	✓	–	post hoc	D
Lin, Wright [19]	object masks	✓	✓	foreground	–
Ours	**none new**	✓	✓	**les., no art.**	**D, E**

III. Method

A. Prototypical part network

We build on a DenseNet-121 backbone [23]. Two 1×1 convolutions with a final sigmoid turn its output into a latent map $z \in [0,1]^{H \times W \times D}$, here with $H = W = 7$ and $D = 128$. The network keeps $m = 20$ prototypes $p_j \in \mathbb{R}^D$, ten for each class, and $c(j)$ denotes the class of prototype j. For every cell we compute the distance $d_j(h,w) = \|z_{hw} - p_j\|_2^2$ and convert it into a similarity $s_j(h,w) = \log \frac{d_j(h,w)+1}{d_j(h,w)+\epsilon}$ with $\epsilon = 10^{-4}$. A prototype is as active as its best-matching cell, $a_j = \max_{h,w} s_j(h,w)$, and the logits are $\mathbf{W}\boldsymbol{a}$. Training minimizes

$$\mathcal{L} = \mathcal{L}_{\mathrm{CE}} + \lambda_c \mathcal{L}_{\mathrm{clst}} - \lambda_s \mathcal{L}_{\mathrm{sep}} + \lambda_l \mathcal{L}_{\mathrm{les}} + \lambda_a \mathcal{L}_{\mathrm{art}}, \quad (1)$$

where $\mathcal{L}_{\mathrm{CE}}$ is a class-weighted cross-entropy and the cluster and separation costs measure the distance to the closest prototype of the own and of the other class [10] ($\lambda_c = 0.8$, $\lambda_s = 0.08$ as originally). The proposed approach consists of the lesion-attraction term $\mathcal{L}_{\mathrm{les}}$ ($\lambda_l = 1$) and the constrained projection of Section III-C; the artifact-suppression term $\mathcal{L}_{\mathrm{art}}$ ($\lambda_a = 1$ in our main runs, 0 in the ablation) is examined in Section VI-H. All three use the masks of Section IV.

B. Lesion attraction (core component)

The masks undergo the same geometric augmentation as the image; average pooling to the 7×7 grid gives, for each cell, the fraction covered by lesion, $M^L(h,w)$, and by artifacts, $M^A(h,w)$. Where a prototype looks is described by its spatial attention $\alpha_j = \mathrm{softmax}_{h,w}(s_j/\tau)$, $\tau = 0.5$, a soft version of the max-pooling location that produces a_j. Each prototype is weighted by $\omega_j = a_j / \sum_k a_k$ (treated as a constant), so that the prototypes the model actually uses count most. With

$L_{ij} = \sum_{h,w} \alpha_{ij} M_i^L$, the attention mass of prototype j on the lesion of image i,

$$\mathcal{L}_{\text{les}} = -\frac{1}{|\mathcal{B}_L|} \sum_{i \in \mathcal{B}_L} \sum_{j:\, c(j)=y_i} \tilde{\omega}_{ij} \log\left(L_{ij} + \delta\right), \qquad (2)$$

where $\tilde{\omega}$ renormalizes ω over the prototypes of the true class y_i, $\delta = 10^{-4}$, and $\mathcal{B}_L$ holds the batch images whose crop still contains lesion. The logarithm penalizes prototypes that almost ignore the lesion much more than those that already look at it, and since α_j sums to one, raising the lesion mass lowers the mass on everything else, artifacts included. Unlike the fine-annotation loss of IAIA-BL [12], [16], which penalizes the magnitude of activation outside a mask, $\mathcal{L}_{\text{les}}$ acts on the normalized attention and leaves the scale of the activations to the classification losses.

C. Lesion- and artifact-constrained projection (core component)

In the "push" step, ProtoPNet replaces each prototype with the closest latent patch from a training image of its class; nothing prevents this patch from lying on a ruler, a black corner or healthy skin, even if the attention was well placed during training. We therefore admit only cells with $M^L \geq 0.5$ and $M^A \leq 0.05$. Every prototype is then, by construction, the latent vector of a cell whose 32×32 input patch is at least half lesion and contains at most 5% detected artifact, so the "that part" of each explanation points to lesion tissue. The constraint fixes where a prototype comes from, not every pixel that shapes its latent vector, since a cell also sees its surroundings (quantified in Section VI-C). After the push, only $\mathbf{W}$ is trained, with backbone, add-on layers and batch-normalization statistics fixed, so each prototype stays identical to its source patch. As five last-layer epochs did not solve this convex problem for *any* prototype model, we re-optimize $\mathbf{W}$ to convergence on fixed training activations (weighted cross-entropy plus an ℓ_1 penalty of 10^{-4} on negative-class connections [10]; 3,000 full-batch Adam steps, cosine-annealed learning rate from 0.02), identically for every prototype model and without validation data, and also report results without it.

IV. Masks: Provenance, Reliability and Leakage Control

Both core components depend on lesion masks; we document their provenance, reliability and use, and why they cannot leak test information.

A. Provenance of the lesion masks

Every image is tagged with its mask source. (i) *Expert masks*: for the 10,015 HAM10000 images (39.5% of ISIC 2019) we use the public lesion segmentations of HAM10000 [24] unchanged. (ii) *Predicted masks*: the 15,316 BCN20000 [25] and MSK [26] images have no public masks, so we train a DeepLabV3 [27] network with an ImageNet-initialized MobileNetV3-Large [28] encoder at 256×256 (12 epochs, AdamW, one-cycle learning rate up to 5×10^{-4}, binary cross-entropy plus soft Dice, random flips and 90° rotations) *only* on the 8,163 HAM10000 images of our training split, which is fixed beforehand. Its prediction is averaged with that of the flipped image, thresholded at 0.5, reduced to the largest connected component and hole-filled. Masks covering less than 0.5% of the image (19 images, 0.08%) are marked invalid and excluded from mask-based losses and metrics.

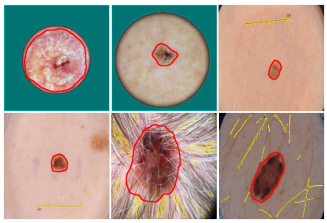


Fig. 1. Automatic masks on ISIC 2019 images (red: lesion; shaded cyan: dark corners, top left and center; yellow: thin dark structures, i.e. rulers, top right and bottom left, and hair, bottom center and right). Artifact masks never overlap the lesion.

B. Validation of the lesion masks

On the 1,852 HAM10000 images of the validation and test splits, unseen by the segmentation network, the predicted masks reach a mean Dice of 0.9445 (median 0.9672, mean Jaccard 0.9015) against the expert masks, and on the 200 expert-segmented PH2 images [29] from another clinic and device, without adaptation, a Dice of 0.918. Section VI-I tests whether our conclusions depend on mask quality.

C. Artifact masks

Artifact masks come from fixed, label-free image operations at 256×256 and always exclude the dilated lesion. *Dark corners* (DC) are dark regions (HSV value < 50) touching the border and at least 8 px thick on average; *thin dark structures* (TS: hair, ruler ticks) are found with a 13×13 black-hat filter as in DullRazor [30] plus an elongation test; *bright structures* (SP: specular highlights, gel-bubble rims, also light hairs) with an 11×11 white top-hat filter; and *ink or colored patches* (INK) as strongly saturated blue-to-violet pixels (Fig. 1). An image is flagged once a mask covers more than 1%, 0.3%, 0.05% or 0.1% of it. *Validation.* Against the public manual labels of Bissoto *et al.* [5], available for 2,593 of our MSK images, the flags are precise but conservative (precision/recall: DC vs. dark corner 0.92/0.10, TS vs. hair or ruler 0.96/0.62, SP vs. gel bubble or border 0.86/0.35, INK vs. ink or patches 0.74/0.25); a visual check of 24 random DC and 24 random TS flags agrees (24 and 20 correct). Many artifacts, above all faint dark corners (37% manual vs. 4% detected prevalence), are missed, so "without artifacts" means "without *detected* artifacts": an undetected artifact can enter a prototype cell, artifact shares count detected artifacts only, and the reference group of Section VI-E contains undetected ones, which pulls A toward 0.5 for every model.

D. How masks are used, and why they cannot leak

Masks enter at exactly four places: the training losses, the constrained projection, the attribution metrics and the placement of counterfactual edits. The classifier never receives

a mask, so its test results do not depend on test masks. Leakage is prevented at every stage:

- *Split first.* The split is computed from image and lesion identifiers and image content only, before any model, including the segmentation network, is trained.
- *Group-level split.* Images are grouped by lesion identifier and by near-duplicate pairs, so re-acquisitions and resized copies of a lesion cannot end up in different sets (Section V-A); grouping look-alike pairs can only remove leakage. An audit finds no shared group, lesion identifier or image between any two sets, and all 342 near-identical pairs (thumbnail correlation ≥ 0.97) and all other verified near-duplicate pairs lie within one set.
- *Training-only supervision.* The segmentation network uses the 8,163 training-split masks only; the 1,852 expert masks of validation and test serve only for evaluation. Mask-based losses and projection use training images only, thresholds and checkpoints are chosen on validation, and the last-layer re-optimization uses training activations only.
- *Label-free masks.* Neither the segmentation network nor the artifact detectors see a diagnosis, so masks cannot carry label information.
- *Untouched external sets.* Derm7pt, PAD-UFES-20 and PH2 are used only for evaluation; removing external images flagged as possible near-duplicates of ISIC training images (8, 6 and 7) changes no external AUC by more than 0.004.

TABLE II
THE ISIC 2019 SPLIT: IMAGES, DIAGNOSES, SOURCES (IMAGES/MELANOMAS) AND PREVALENCE OF DETECTED ARTIFACTS. HAM10000 IMAGES HAVE EXPERT LESION MASKS, BCN20000 AND MSK IMAGES PREDICTED ONES (VALID: 99.91, 99.96 AND 100.0%). THRESHOLDS AND CHECKPOINTS ARE SELECTED ON VALIDATION. EXTERNAL TEST SETS: DERM7PT, 1,011 CASES WITH A DERMOSCOPIC AND A CLINICAL IMAGE EACH (252 MELANOMAS); PAD-UFES-20, 2,298 IMAGES (52 MELANOMAS); PH2, 200 IMAGES (SEGMENTATION ONLY).

	Train	Val.	Test	All
Images	20,552	2,459	2,320	25,331
Melanoma rate (%)	17.8	19.3	16.8	17.9
Diagnosis				
Melanoma	3,657	475	390	4,522
Nevus	10,523	1,208	1,144	12,875
Basal cell carc.	2,671	322	330	3,323
Actinic keratosis	693	80	94	867
Benign keratosis	2,095	278	251	2,624
Dermatofibroma	193	16	30	239
Vascular lesion	201	27	25	253
Squamous cell carc.	519	53	56	628
Source (images / melanomas)				
BCN20000	10,037/2,299	1,225/310	1,151/248	12,413/2,857
HAM10000	8,163/917	952/103	900/93	10,015/1,113
MSK	2,352/441	282/62	269/49	2,903/552
Artifact prevalence (%)				
Dark corners	25.5	25.7	25.1	–
Thin dark structures	24.2	25.7	23.4	–
Bright structures	13.8	14.4	13.3	–
Ink / patches	1.6	1.9	2.0	–

V. EXPERIMENTAL SETUP

A. Data and leakage-free split

The task is melanoma versus all other diagnoses on the 25,331 training images of ISIC 2019 (HAM10000 [24], BCN20000 [25], MSK [26]; 4,522 melanomas, 17.9%), resized to 256×256. Unlike common practice, we keep the 2,084 MSK images without lesion identifier. To avoid leakage we group images by lesion identifier and by near-duplicate pairs (perceptual-hash distance ≤ 8 and thumbnail correlation ≥ 0.9 for the full and the central image); the 13,259 groups are split into ten stratified folds, one for testing and one for validation (about 81/10/9% per source), with almost identical composition (Table II). External test sets are Derm7pt [31], with a dermoscopic and a clinical image for each of 1,011 cases (melanomas include *in situ* lesions and metastases), and the smartphone images of PAD-UFES-20 [32]; PH2 has no diagnoses in our copy and serves only to check the segmentation.

B. Models and training

All configurations share backbone (ImageNet-pretrained DenseNet-121), input size (224×224), data, split and schedule. **CNN**: backbone, global average pooling, linear layer (black box). **PPN-std**: ProtoPNet with standard augmentation (random crop, horizontal flip). **PPN**: ProtoPNet with strong augmentation (random resized crop, scale $[0.4, 1]$; horizontal and vertical flips; $\pm 20\%$ brightness, contrast, saturation), used by all following models. **PPN-FA**: PPN with the IAIA-BL fine-annotation loss [16] on our lesion masks as in Correia *et al.* [12], with their $\lambda = 10^{-3}$ but computed on the 7×7 maps and averaged over the batch, which weakens it (original formulations and larger weights: Section VI-G). **Ours**: lesion attraction and constrained projection; this main model also contains the exploratory $\mathcal{L}_{\mathrm{art}}$ ($\lambda_l = \lambda_a = 1$ in Eq. (1)), whereas the two-component method alone is Ours-noArt. **Ablations**: without $\mathcal{L}_{\mathrm{les}}$ (Ours-noLes), without $\mathcal{L}_{\mathrm{art}}$ (Ours-noArt, the two core components only), with the standard projection (Ours-stdPush).

We train with Adam, batch size 64 and mixed precision: 5 epochs with frozen backbone (add-on layers and prototypes at 3×10^{-3}), 15 epochs of all layers (backbone 10^{-4}, weight decay 10^{-3}, learning rate divided by 10 every 5 epochs), then the projection and 5 last-layer epochs; the CNN is trained for the same 20 epochs. Class weights are 0.608 and 2.81. The CNN keeps its best validation checkpoint; prototype models are always evaluated after the projection, with the last-layer epoch chosen on validation. Each configuration runs with seeds 0, 1 and 2 (about 30 minutes per run on an NVIDIA T4). *Resolutions.* Images and masks are stored at 256×256 and receive the same random crop and flips before bilinear resampling to the 224×224 input (test: plain resize); pooled to the 7×7 grid, each cell covers a 32×32 input patch, and attribution maps are compared with masks at 224×224. All loss weights were fixed in advance and not tuned on test data.

C. Evaluation protocol

Classification. Each run's threshold is fixed on ISIC validation (Youden index) and reused for every test set. AUC, balanced accuracy (BAcc), sensitivity and specificity are mean $\pm$ std over seeds; tests use seed ensembles (mean probability of the three seeds).

Attribution. For a prototype model, the similarity maps of the three most activated prototypes are upsampled to 224×224, and their top 5% of pixels (the 95th-percentile region that ProtoPNet visualizes) form the hot region; for the CNN we use the class-activation map, which with a global-average-pooling head equals Grad-CAM [33] up to a positive factor [34]. We report the share of the hot region on the lesion and on artifacts, whether its maximum falls on an artifact (pointing), and the *enrichment*, artifact share over artifact area (above 1: more attention than the area explains).

Counterfactuals. Each test image is modified in four ways outside a 4-pixel margin around the lesion: a ruler along one border, 12–29 curved hairs, a dark vignette, or 2–4 purple ink arcs. Conversely, detected thin dark structures (real hair and ruler marks, 544 images) are erased by Telea inpainting [35] outside a 2-pixel margin; as a null control, a region of the same shape is shifted onto artifact-free skin and inpainted. Every lesion pixel is verified to stay bit-identical. We record the mean $|\Delta p|$ of the melanoma probability and the rate of flipped decisions and, for the three prototypes most active on the original image, the Spearman correlation ρ of the top-1 map before and after and the *capture*, the part of the new hot region on inserted pixels.

Statistics. The 2,320 test images form 1,234 groups of the split (a lesion with its near-duplicates; 471 with several images), so we resample whole groups: ensemble AUCs get 95% CIs from 1,000 group resamples and are compared by a paired bootstrap with 2,000 (two-sided p), and per-image metrics, averaged over seeds and then within groups, by the paired Wilcoxon test. Derm7pt (one image per case) uses the paired DeLong test [36], [37], PAD-UFES-20 a bootstrap over lesions. The 12 tests behind the main claims (Ours vs. each of PPN-std, PPN and PPN-FA on AUC, artifact share and flip rate under inserted dark corners and rulers) form the primary family, corrected with Holm's method; all other tests are secondary and uncorrected. Image-level tests, which ignore the grouping, agree unless noted. All tests treat the trained models as fixed and capture the uncertainty of the test sample; variation over training runs is shown by the std over the three seeds.

VI. Results

We compare the five main models (CNN, PPN-std, PPN, PPN-FA, Ours) unless stated otherwise; ablations, mask-loss variants and the artifact-suppression term follow in Sections VI-F–VI-H. Bold marks the best of the four main prototype models.

TABLE III
ARTIFACT–MELANOMA ASSOCIATION (TRAINING SPLIT). OR: ODDS RATIO; MH: MANTEL–HAENSZEL OR STRATIFIED BY SOURCE.

Artifact	Prev.	$P(\mathrm{m}\|\mathrm{a})$	$P(\mathrm{m}\|\neg\mathrm{a})$	OR	MH OR [95% CI]
Dark corners	25.5%	.263	.149	2.03	1.555 [1.43, 1.69]
Bright structures	13.8%	.266	.164	1.85	1.751 [1.59, 1.92]
Thin dark struct.	24.2%	.174	.179	0.97	0.988 [0.90, 1.08]
Ink / patches	1.6%	.202	.178	1.18	1.114 [0.85, 1.46]
Any	48.8%	.210	.148	1.53	1.265 [1.17, 1.36]

A. Artifacts are associated with melanoma

In the training split, melanomas show dark corners and bright structures significantly more often than other lesions (Table III). Source could explain this (dark corners: 47% of BCN vs. 5% of HAM images, and BCN has the highest melanoma rate), but stratifying by source does not remove it: the Mantel–Haenszel odds ratios stay clearly above 1, and for dark corners the odds ratio exceeds 1 in every source (BCN 1.41, HAM 3.10, MSK 2.24), for bright structures in BCN and HAM (2.11, 1.92) but not in MSK (0.88). The strength differs between sources (Breslow–Day $p < 10^{-9}$), so pooled odds ratios are averages. Thin dark structures and ink show no association.

B. Classification

On the ISIC test set, Ours reaches the highest AUC among the prototype models (0.865 ± 0.006, ensemble 0.884; Table IV); its ensemble exceeds those of PPN-std, PPN and PPN-FA by 0.022, 0.018 and 0.017 (95% CI $[0.009, 0.036]$, $[0.004, 0.032]$, $[0.005, 0.031]$; Holm-adjusted $p = 0.006$, 0.014, 0.014), its seeds vary less (std 0.006 vs. 0.040 for PPN), and each Ours seed beats each of the nine baseline seeds (worst 0.8592 vs. best 0.8526; on Derm7pt-dermoscopic 0.7405 vs. 0.7343), a descriptive check with three seeds per model. At the validation threshold, Ours trades some sensitivity for specificity (Table IV). The black-box CNN remains more accurate (0.912 vs. 0.884, $p = 0.009$), a price of about 3 AUC points for interpretability (4.6 for PPN). Without the last-layer re-optimization (five-epoch last layer, epoch chosen on validation) all prototype models score lower, but Ours stays best (0.846 ± 0.007 vs. 0.797 ± 0.027, 0.770 ± 0.027 and 0.765 ± 0.023; Ours-noLes, Ours-noArt and Ours-stdPush 0.787 ± 0.045, 0.834 ± 0.029 and 0.841 ± 0.002), significantly so ($p \leq 0.001$), and loses least (0.019 vs. 0.039–0.079).

Externally, Ours has the highest AUC of the four main prototype models on Derm7pt-dermoscopic (0.749) and Derm7pt-clinical (0.657). On the dermoscopic images its ensemble beats PPN-FA and PPN-std ($p = 1.7 \times 10^{-5}$, 2.8×10^{-6}, also after Holm correction) but not PPN ($p = 0.20$, high seed variance); on the clinical images its advantage over PPN-std and PPN ($p = 0.023$, 0.046) does not survive Holm correction ($p = 0.069$, 0.092), and over PPN-FA it is not significant ($p = 0.30$). On PAD-UFES-20 (52 melanoma images of 36 lesions), Ours beats PPN-std ($p = 0.001$) and does not differ from PPN and PPN-FA ($p \geq 0.23$). The black-box EPVT reports 0.837 on Derm7pt-dermoscopic [7], but for melanoma

TABLE IV

CLASSIFICATION (MEAN ± STD OVER 3 SEEDS; THRESHOLD FROM ISIC VALIDATION). ENS.: SEED-ENSEMBLE AUC WITH 95% CI; p: ENSEMBLE AGAINST OURS ON ISIC TEST (PAIRED BOOTSTRAP OVER LESION GROUPS). OURS: TWO CORE COMPONENTS AND THE EXPLORATORY $\mathcal{L}_{\text{art}}$; OURS-NOART: THE TWO CORE COMPONENTS ALONE.

	ISIC 2019 test (n = 2,320)						External AUC		
Model	AUC	BAcc	Sens.	Spec.	Ens. AUC [95% CI]	p	Derm7pt-derm	Derm7pt-clin	PAD-UFES-20
CNN (black box)	.902±.006	.814±.011	.808±.044	.820±.040	.912 [.884, .935]	.009	.799±.023	.704±.022	.721±.022
PPN-std	.835±.013	.752±.014	.769±.063	.734±.040	.861 [.827, .888]	.002	.677±.027	.624±.023	.587±.019
PPN	.825±.040	.758±.010	**.786±.068**	.730±.087	.866 [.830, .894]	.014	.688±.072	.618±.038	**.665±.016**
PPN-FA (IAIA-BL)	.844±.008	.744±.021	.689±.045	.800±.005	.866 [.830, .895]	.007	.696±.024	.640±.013	.630±.059
Ours	**.865±.006**	**.780±.014**	.734±.075	**.825±.046**	**.884 [.850, .910]**	–	**.749±.009**	**.657±.019**	.647±.108
Ours-noLes	.830±.015	.746±.030	.766±.091	.726±.032	.856 [.821, .885]	<.001	.717±.016	.636±.033	.628±.047
Ours-noArt	.859±.011	.772±.003	.733±.056	.811±.054	.877 [.845, .904]	.22	.752±.014	.637±.030	.678±.018
Ours-stdPush	.863±.005	.785±.002	.823±.043	.747±.047	.884 [.852, .911]	.86	.747±.005	.670±.019	.658±.108

TABLE V

WHERE THE MODELS LOOK (ISIC TEST, MEAN ± STD OVER SEEDS). ART.: HOT-REGION FRACTION ON ARTIFACTS IN IMAGES WITH ARTIFACTS; ENR.: ARTIFACT ENRICHMENT; POINT: MAXIMUM ON AN ARTIFACT; L-PR.: PROTOTYPES PROJECTED ONTO LESION CELLS (OF 20).

Model	Lesion↑	Art.↓	Enr.↓	Point↓	L-pr.
CNN	.466±.040	.231±.024	1.52±.46	.117±.007	–
PPN-std	.518±.036	.082±.049	0.54±.31	.050±.029	12.0
PPN	.562±.012	.066±.045	0.46±.04	.042±.040	14.3
PPN-FA	.530±.051	.102±.024	0.68±.05	.050±.006	10.0
Ours	**.589±.040**	**.048±.015**	**0.37±.13**	**.013±.005**	**20.0**
Ours-noLes	.565±.051	.106±.056	0.67±.11	.053±.029	20.0
Ours-noArt	.594±.028	.042±.008	0.40±.03	.018±.007	20.0
Ours-stdPush	.515±.063	.086±.052	0.58±.30	.034±.024	7.7

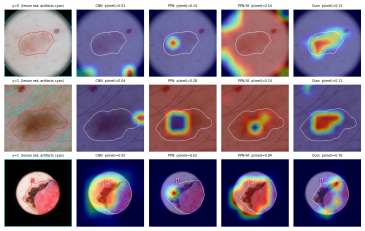


Fig. 2. ISIC test images with artifacts (left: lesion red, detected artifacts cyan). CNN: class-activation map; prototype models: map of the most activated prototype; p: melanoma probability (seed-0 models). Here the CNN and PPN-FA respond to dark corners and image borders, whereas the top prototype of Ours stays on the lesion.

versus benign, another split and a larger ViT-B/16 backbone; our CNN reaches 0.799.

C. Where the models look

The black-box CNN is the model most drawn to artifacts (Table V, Fig. 2): in images with artifacts 23.1% of its hot region lies on them, 1.5 times their area share, and in 11.7% of the images its maximum sits on an artifact. The prototype models look at artifacts less than their area suggests, Ours least: 4.8% of the hot region (PPN 6.6%, PPN-FA 10.2%), an artifact maximum in 1.3% of images (PPN 4.2%), and the largest lesion fraction (58.9%). All 20 prototypes of every Ours run are lesion patches, whereas PPN-FA projects on average 10.0 onto lesion and 2.0 onto artifact cells (Fig. 3). Most attention on artifacts falls on dark corners, where Ours has the smallest share (Table VI). All differences between Ours and each other main model in lesion share, artifact share, enrichment and pointing are significant (Wilcoxon on lesion-group means, $p \leq 5.6 \times 10^{-4}$; artifact shares in the primary family: Holm-adjusted $p \leq 1.3 \times 10^{-5}$). On Derm7pt-dermoscopic, Ours again has the lowest artifact share, enrichment and pointing rate (1.5% vs. 4.4% for PPN and 15.3% for the CNN; $p \leq 8.6 \times 10^{-3}$), but the three prototype baselines place more of their hot region on the lesion (0.75–0.79 vs. 0.73; $p \leq 1.2 \times 10^{-4}$).

Receptive field of the prototypes. We computed the effective receptive field (ERF) of the source cell of every prototype, the input-gradient magnitude of its latent vector on the source image (eight random sign projections of the 128 channels). For every model only 8–12% of the ERF mass lies inside the cell, and half of it spreads over 8–11% of the image. For Ours, $69 \pm 3\%$ lies on the lesion (10th percentile over prototypes 42%), against 48–53% for PPN-std, PPN and PPN-FA (10th percentiles 11–21%), and 2.5% on detected artifacts (4.3%, 2.7%, 7.3%). This stems from where prototypes are taken, not from a narrower field: the constraint selects source images whose lesion covers 54% of the image (baselines 40%), while relative to the lesion area all models concentrate the ERF on the lesion by a similar median factor (1.2–1.3). Without the constraint (Ours-stdPush) the lesion share drops to 47%, whereas Ours-noLes and Ours-noArt keep 71–72%. For 13% of the prototypes of Ours (baselines 22–28%) the ERF draws less on the lesion than its area would, so anchoring covers the projected cell and most, not all, of its receptive field (Fig. 4).

D. Counterfactual robustness

Table VII summarizes the interventions (2,063–2,318 modified images per insertion type, lesion pixels verified unchanged). Adding dark corners *increases* the melanoma probability of every model (mean signed change $+0.041$ CNN, $+0.048$ PPN, $+0.020$ Ours), matching the association in Table III: the models have absorbed the shortcut, Ours least. Ours roughly halves the flip rate of PPN for dark corners (10.9%→5.2%) and rulers (7.5%→3.6%), reduces it for ink (9.5%→5.6%) and hair (13.9%→9.2%), and flips fewer decisions than the CNN for all four insertions. These reductions

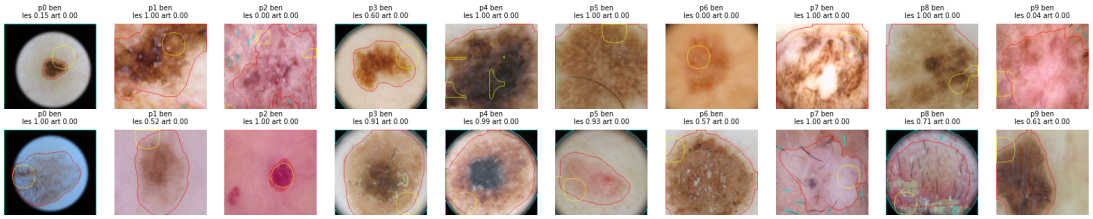


Fig. 3. Source patches of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0 ("les"/"art": lesion/artifact fraction of the projected cell; red: lesion; cyan: artifacts; yellow: 95th-percentile region on the source image). Four PPN prototypes come from background skin (lesion fraction 0.00–0.15); all of Ours are lesion cells (≥ 0.52), several at the lesion border.

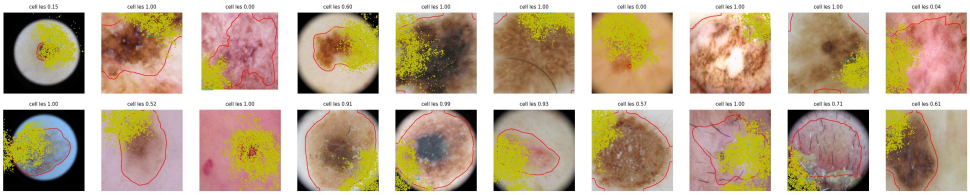


Fig. 4. Effective receptive fields of the ten benign-class prototypes of PPN (top) and Ours (bottom), seed 0, on their source images (red: lesion; cyan: projected cell; yellow: smallest region holding half of the ERF; "cell les": lesion fraction of the cell). PPN prototypes from background cells draw on skin and vignette; those of Ours draw mostly, though not only, on the lesion.

TABLE VI
ATTRIBUTION BY ARTIFACT TYPE (ISIC TEST; HOT-REGION SHARE ON THE GIVEN ARTIFACT IN IMAGES THAT CONTAIN IT) AND ON THE EXTERNAL DERM7PT-DERMOSCOPIC SET (MEAN OVER 3 SEEDS).

	ISIC test, by type				Derm7pt-derm		
Model	DC	TS	SP	INK	Lesion	Point	Enr.
CNN	.365	.016	.017	.029	.605	.153	.63
PPN-std	.123	**.010**	.015	.049	.749	.075	.25
PPN	.098	**.010**	.012	.036	**.789**	.044	.24
PPN-FA	.158	**.010**	**.010**	.056	.760	.077	.28
Ours	**.067**	.011	.016	**.032**	.734	**.015**	**.18**
Ours-noLes	.162	.011	.013	.044	.772	.057	.30
Ours-noArt	.056	.012	.020	.040	.713	.017	.23
Ours-stdPush	.129	.012	.019	.030	.633	.033	.35

are significant against every other main model for all four insertions (Wilcoxon on lesion-group means, $p \leq 4.9 \times 10^{-8}$ against the prototype baselines, Holm-adjusted $p \leq 2.5 \times 10^{-7}$ for dark corners and rulers; $p \leq 0.008$ against the CNN), as is the smaller $|\Delta p|$ against PPN ($p < 10^{-27}$) and, except for hair ($p = 0.09$), the CNN ($p \leq 2.3 \times 10^{-4}$). Removing real hair and ruler marks changes more decisions than the null control for every model (Ours 4.4% vs. 1.6%), so the sensitivity stems from the artifacts, not from inpainting.

Its explanations also move the least (Fig. 5): after adding dark corners its top-1 map keeps $\rho = 0.92$ (PPN 0.68), and only 3.4% of its hot region moves onto the inserted pixels (PPN 6.1%, PPN-FA 11.4%; $p < 10^{-84}$). The exception is ink, drawn just outside the lesion border where lesion-anchored prototypes respond: 1.8% of the hot region of Ours falls on it vs. 1.0% for PPN, although Ours still flips fewer decisions.

E. Shortcuts in unedited images

As a complementary test without editing, we ask whether artifacts present in the test images raise the melanoma score of *benign* lesions, measured by the threshold-free probability A

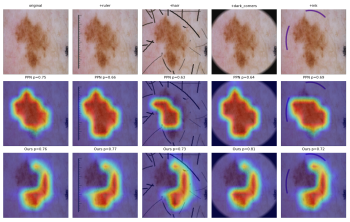


Fig. 5. Counterfactual example with unchanged lesion pixels (second of the two test images drawn by the evaluation notebook; the first is in the repository). Rows 2–3: map of the prototype most active on the original image and melanoma probability p for PPN and Ours; the map of Ours keeps its place under all four insertions, that of PPN shifts when hair is added.

of Table VIII ($A = 0.5$: no effect). On ISIC, dark corners double the false-positive rate of PPN-FA (16%→33%) but raise that of Ours only from 17% to 20%. The effect A of Ours (0.57) is significantly smaller than that of PPN, PPN-FA and Ours-noLes (0.61–0.66; $p < 0.001$) and not significantly different from that of PPN-std and the CNN ($p \geq 0.77$): our weak mask loss leaves the real dark-corner shortcut in place, lesion attraction keeps it small, and the original mask-loss formulations reduce it as well (Section VI-G). The same holds for bright structures, where Ours has the smallest effect of the prototype models (0.59 vs. 0.60–0.62; $p \leq 0.038$ against PPN-std and PPN-FA), and most clearly for dark border regions in the clinical Derm7pt photographs (0.69 vs. 0.78–0.82; DeLong $p \leq 1.4 \times 10^{-6}$). The CNN shows a similar or smaller effect, although it attends to dark corners most.

F. Ablation of the two core components

Lesion attraction is the term behind the accuracy gain: without it (Ours-noLes) the test AUC falls to 0.830 ($p < 0.001$), the artifact share of the hot region more than doubles (4.8%→10.6%), and the flip rates for dark corners, rulers and ink roughly double. Since Ours-noLes still uses the constrained projection, all of its prototypes are lesion patches: a correct

TABLE VII

COUNTERFACTUAL INTERVENTIONS ON ISIC TEST (LESION PIXELS UNCHANGED; MEAN OVER 3 SEEDS). FLIP: DECISION CHANGES (%); $|\Delta p|$: MEAN ABSOLUTE CHANGE OF THE MELANOMA PROBABILITY; ρ: SPEARMAN CORRELATION OF THE TOP-1 PROTOTYPE MAP BEFORE/AFTER; CAP.: SHARE OF THE NEW HOT REGION ON INSERTED PIXELS (%). THE NULL CONTROL IS NOT RANKED.

	+Dark corners		+Ruler		+Hair		+Ink		Remove hair/ruler		Null control		Expl.: dark corners		Expl.: ruler	
Model	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	Flip	$\|\Delta p\|$	$\rho\uparrow$	Cap.$\downarrow$	$\rho\uparrow$	Cap.$\downarrow$
CNN	6.1	.049	5.6	.049	12.5	.104	9.7	.080	4.7	.036	1.0	.008	–	–	–	–
PPN-std	10.9	.060	8.3	.041	14.9	.092	9.9	.058	8.8	.036	4.3	.013	.635	7.1	.773	1.5
PPN	10.9	.067	7.5	.045	13.9	.096	9.5	.062	7.4	.045	3.6	.019	.680	6.1	.804	1.1
PPN-FA	10.6	.049	6.5	.028	15.1	.084	9.1	.046	7.2	**.030**	3.2	.010	.714	11.4	.874	1.1
Ours	**5.2**	**.035**	**3.6**	**.025**	**9.2**	**.081**	**5.6**	**.045**	**4.4**	.032	1.6	.011	**.918**	**3.4**	**.952**	**0.7**
Ours-noLes	10.2	.053	8.7	.040	14.3	.083	10.1	.049	8.3	.042	3.9	.016	.705	8.6	.827	1.7
Ours-noArt	5.4	.036	3.5	.023	8.7	.061	6.1	.041	4.4	.027	1.0	.011	.884	3.4	.949	1.0
Ours-stdPush	4.5	.034	4.1	.028	9.6	.077	5.8	.045	4.5	.033	1.9	.011	.923	7.5	.944	1.4

TABLE VIII

REAL, UNEDITED ARTIFACTS ON BENIGN TEST LESIONS. A: PROBABILITY THAT A BENIGN LESION WITH THE ARTIFACT SCORES HIGHER THAN ONE WITHOUT IT (SEED ENSEMBLES; 0.5: NO EFFECT); FPR: FALSE-POSITIVE RATE (%) WITH/WITHOUT THE ARTIFACT (VALIDATION THRESHOLD, MEAN OVER SEEDS). BENIGN IMAGES WITH/WITHOUT: 447/1,483 (ISIC, DC), 251/1,679 (ISIC, SP), 515/244 (DERM7PT-CLINICAL, DC).

	ISIC, dark corners		ISIC, bright struct.		Derm7pt-clin., DC	
Model	$A\downarrow$	FPR	$A\downarrow$	FPR	$A\downarrow$	FPR
CNN	.57	22/17	.58	19/18	.59	18/9
PPN-std	.58	31/25	.61	32/26	.82	60/21
PPN	.61	30/26	.60	33/26	.78	35/11
PPN-FA	.66	33/16	.62	26/19	.80	46/15
Ours	**.57**	**20/17**	**.59**	**19/17**	**.69**	**24/12**
Ours-noLes	.66	37/25	.63	37/26	.77	50/24
Ours-noArt	.58	20/19	.62	25/18	.75	37/12
Ours-stdPush	.57	28/25	.59	29/25	.65	33/19

TABLE IX

MASK-LOSS VARIANTS (ISIC TEST UNLESS NOTED; MEAN OVER 3 SEEDS). PPN-FA: 7×7 MAPS, BATCH MEAN; CORREIA [12], IAIA-BL [16]: 224×224 MAPS, BATCH SUM, $\lambda=10^{-3}$. D7-D: DERM7PT-DERMOSCOPIC AUC; L-PR.: LESION PROTOTYPES (OF 20); ART.: HOT-REGION SHARE ON ARTIFACTS; FLIP, ρ: INSERTED DARK CORNERS; A: REAL DARK CORNERS.

Mask loss	AUC	D7-d	L-pr.	Art.$\downarrow$	Flip$\downarrow$	$\rho\uparrow$	$A\downarrow$
None (PPN)	.825±.040	.688	14.3	.066	10.9	.68	.61
PPN-FA, 10^{-3}	.844±.008	.696	10.0	.102	10.6	.71	.66
PPN-FA, 10^{-2}	.833±.018	.695	16.0	.121	12.2	.80	.66
PPN-FA, 10^{-1}	.849±.006	.711	9.3	.061	7.2	.84	.60
Correia	**.867±.011**	**.771**	13.0	.141	6.5	.87	**.57**
IAIA-BL	.845±.011	.715	18.0	**.039**	6.2	.88	.58
Ours (no mask loss)	.865±.006	.749	**20.0**	.048	**5.2**	**.92**	**.57**

projection alone is not enough, the attention during training has to be anchored as well.

The constrained projection does not change the accuracy measurably (Ours-stdPush; ensemble difference within $[-0.006, 0.005]$, 95% CI) but determines what the prototypes *are*: with the standard projection only 7.7 of 20 prototypes lie on the lesion, the artifact share of the hot region rises from 4.8% to 8.6%, and the share captured by added dark corners from 3.4% to 7.5% ($p < 10^{-148}$). Accuracy alone would hide this difference.

G. Strength of the mask-loss baseline

PPN-FA computes the fine-annotation loss on the 7×7 maps and averages it over the batch; IAIA-BL [16] and Correia *et al.* [12] compute it at 224×224 and sum it over the batch, which makes the same $\lambda=10^{-3}$ about two orders of magnitude stronger (Correia *et al.* penalize own-class prototypes only). We therefore trained both original formulations and ours with $\lambda=10^{-2}$ and 10^{-1}, three seeds each (Table IX; secondary tests). The ISIC AUC of the Correia formulation does not differ significantly from that of Ours (ensemble difference -0.001, 95% CI $[-0.012, 0.011]$), and it generalizes better (Derm7pt-dermoscopic 0.771 vs. 0.749, Derm7pt-clinical 0.702 vs. 0.657, PAD-UFES-20 0.699 vs. 0.647; $p \le 0.028$), yet only 13 of its 20 prototypes are lesion cells and its hot region lies three times more on artifacts (14.1% vs. 4.8%, $p < 10^{-29}$). The IAIA-BL formulation keeps its hot region on the lesion (83%) and off artifacts (3.9%) better than Ours ($p < 0.002$) but is less accurate (0.845; every Ours seed is higher, ensembles $p = 0.057$) and leaves 2 of 20 prototypes off the lesion. Every variant flips more decisions than Ours under inserted dark corners (6.2–12.2% vs. 5.2%) and ink (6.6–10.7% vs. 5.6%; $p \le 0.02$); for rulers only the original formulations ($p \ge 0.052$), and for hair only the Correia one ($p = 0.37$), do not differ significantly from Ours, and none keeps its explanation as stable under dark corners (ρ 0.71–0.88 vs. 0.92). The real dark-corner effect of both original formulations does not differ significantly from that of Ours (A 0.57–0.58 vs. 0.57, $p \ge 0.26$), unlike our weak version (0.66). Mask supervision thus shapes where the network looks and helps it generalize, while anchoring fixes what the prototypes are and keeps decisions stable: the two are complementary.

H. Artifact suppression: a largely negative result

We also explored a term that penalizes attention on artifacts directly; with $A_{ij} = \sum_{h,w} \alpha_{ij} M_i^A$, the attention mass of prototype j on the artifacts of image i,

$$\mathcal{L}_{\text{art}} = \frac{1}{|\mathcal{B}|} \sum_{i\in\mathcal{B}} \sum_{j=1}^{m} \omega_{ij} A_{ij}. \tag{3}$$

TABLE X

ROBUSTNESS TO MASK QUALITY (ISIC TEST, MEAN ± STD OVER 3 SEEDS): ATTRIBUTION AS IN TABLE V ON IMAGES WITH EXPERT (HAM10000, $n = 900$) AND PREDICTED MASKS (BCN20000, MSK, $n = 1{,}420$); LESION FRACTION WITH ALL LESION MASKS ERODED OR DILATED BY 5 PX; AUC PER SOURCE (n/MELANOMAS: BCN 1,151/248, HAM 900/93, MSK 269/49).

Model	Expert masks (HAM10000)			Predicted masks (BCN, MSK)			Lesion, perturbed mask		AUC by source		
	Lesion↑	Art.↓	Point↓	Lesion↑	Art.↓	Point↓	−5 px	+5 px	BCN	HAM	MSK
CNN	.444±.044	.043±.014	.015±.006	.480±.038	.277±.026	.182±.010	.434±.038	.496±.040	.883±.012	.934±.003	.861±.002
PPN-std	.416±.085	.016±.005	.004±.002	.582±.046	.098±.059	.079±.047	.480±.036	.552±.035	.825±.018	.880±.016	.711±.022
PPN	.446±.053	.014±.001	.005±.005	**.635±.015**	.079±.056	.065±.062	.525±.013	.595±.012	.819±.051	.860±.034	.692±.016
PPN-FA	.423±.045	.019±.003	.004±.002	.599±.060	.123±.030	.080±.011	.491±.055	.567±.047	.834±.009	.896±.004	.699±.014
Ours	**.534±.062**	**.013±.005**	**.002±.001**	.624±.026	**.056±.018**	**.021±.008**	**.534±.039**	**.640±.041**	**.852±.008**	**.906±.006**	**.761±.018**
Ours-noLes	.503±.089	.023±.013	.007±.006	.603±.064	.126±.071	.082±.049	.525±.055	.601±.047	.815±.017	.879±.013	.704±.032
Ours-noArt	.543±.030	.013±.003	.003±.001	.626±.043	.050±.010	.028±.010	.541±.041	.644±.023	.853±.013	.893±.008	.737±.023
Ours-stdPush	.459±.043	.018±.007	.003±.001	.550±.075	.103±.063	.054±.039	.466±.060	.561±.065	.852±.009	.904±.004	.749±.005

Removing it (Ours-noArt) changes the accuracy significantly neither on ISIC (ensemble difference 0.007, 95% CI $[-0.004, 0.018]$) nor on any external set ($p \geq 0.058$); several counterfactual measures are even better without it (e.g., $|\Delta p|$ 0.061 vs. 0.081 for added hair, $p < 10^{-40}$), and the attribution changes little (artifact share 4.2% vs. 4.8%, $p = 0.36$; pointing 1.8% vs. 1.3%, $p = 5.7 \times 10^{-5}$). The two core components alone carry the explanation gains: against PPN-std, PPN and PPN-FA, Ours-noArt has more hot region on the lesion and less on artifacts and flips fewer decisions under all four insertions ($p \leq 5.2 \times 10^{-4}$), but its AUC gain over them (0.010–0.016) is not significant with lesion groups resampled ($p = 0.073$–0.17; image-level DeLong $p = 0.005$ against PPN-std), so the significant accuracy gain belongs to the main model. $\mathcal{L}_{\text{art}}$ mainly weakens the response to real bright structures and to dark border regions in clinical images (A 0.59 vs. 0.62 and 0.69 vs. 0.75; $p < 0.001$, Table VIII); otherwise its effect is small, because $\mathcal{L}_{\text{les}}$ already lowers the attention on artifacts and the constrained projection removes the remaining route to an artifact prototype. We therefore do not count $\mathcal{L}_{\text{art}}$ as part of the method.

I. Robustness to mask quality

Predicted masks could bias the results if a model looked "on the lesion" only by the measure of an imperfect mask, or if the gains appeared only where masks were predicted. We therefore split the ISIC test set by mask provenance (900 HAM10000 images with expert masks, 1,420 BCN20000 and MSK images with predicted ones) and perturb the lesion masks (Table X).

Expert masks alone. On the HAM10000 test images, Ours places 53.4% of its hot region on the lesion, against 41.6–44.6% for the prototype baselines and 44.4% for the CNN; its advantage is thus *larger* on expert masks (+8.8 points over the best baseline) than on the whole test set (+2.7), so it is not produced by errors of the predicted masks, and its artifact scores are the lowest there too. On the predicted-mask images Ours has the lowest artifact share (5.6% vs. 7.9–27.7%), although PPN places slightly more on the lesion (63.5% vs. 62.4%, within one std).

Perturbed masks. Eroding or dilating every lesion mask by 5 pixels (at 256×256) shifts the lesion fractions of all models alike; the order of the five main models stays the same (Ours > PPN > PPN-FA > PPN-std > CNN; margin of Ours over PPN 0.9 points under erosion, 4.5 under dilation), and artifact share, pointing and enrichment change by at most 0.011.

Classification per source. Ours is the most accurate prototype model in every source, with expert training masks (HAM10000: 0.906 vs. 0.896 for PPN-FA) and with predicted ones (BCN20000: 0.852 vs. 0.834; MSK: 0.761 vs. 0.711 for PPN-std).

Counterfactuals per provenance. A too small predicted mask could let inserted artifacts touch the true lesion, but this does not drive the results: on the expert-mask subset, where the protected region is exact, Ours flips the fewest decisions (dark corners 4.4% vs. 7.1% for the CNN and 10.0–10.3% for the prototype baselines; rulers 2.3% vs. 3.9% and 4.7–6.1%), the predicted-mask subset gives the same picture (dark corners 5.7% vs. 10.7–11.5%, CNN 5.5%; rulers 4.4% vs. 6.7% and 7.7–9.7%), and its $|\Delta p|$ is the smallest in both (0.035–0.036 for dark corners vs. 0.043–0.068). The conclusions thus hold on expert masks alone, under ±5-pixel mask errors and in every source.

VII. DISCUSSION AND LIMITATIONS

The two components play different roles: lesion attraction changes what the network learns and makes it both more accurate and less dependent on artifacts, whereas the constrained projection changes what it shows to the user.

Dependence on masks. Lesion masks are needed only in training, come from public expert segmentations of one source (no new annotation for a new dermoscopic dataset), and enter only as coarse 7×7 cell fractions, which tolerate boundary errors (Section VI-I).

Limitations. (i) A gap of about 3 AUC points to the black-box CNN remains; Correia *et al.* [12] found prototype models on par with CNNs, but on ISIC validation data and after choosing the best of many configurations. (ii) For BCN20000 and MSK no expert masks exist, so their predicted masks are validated only indirectly (held-out HAM10000 images, PH2, subset analysis), and the artifact detectors miss many artifacts (Section IV), so our evidence concerns detected and inserted artifacts, not every artifact in real images. (iii) Inserted artifacts are synthetic; the removal of real artifacts, a null control and the test on unedited images mitigate this, although artifact

effects in unedited images may be partly confounded by lesion appearance. (iv) The 7×7 grid limits the spatial resolution of the explanations, and class-activation and prototype maps differ in kind, so attribution comparisons between the CNN and the prototype models are only indicative. (v) External gains are significant against two of the three main prototype baselines on Derm7pt-dermoscopic, against none after Holm correction on Derm7pt-clinical and against one on PAD-UFES-20, and the original mask loss generalizes better (Section VI-G); one backbone and one prototype architecture were studied, as required for a controlled comparison, although both components apply to any prototype network with similarity maps and a projection step; the weights of our own losses were not varied. (vi) Training variability rests on three seeds per configuration. (vii) This retrospective study does not establish clinical readiness, which would require prospective validation in other populations.

VIII. Conclusion

Classifiers trained on ISIC 2019 exploit artifact shortcuts, and a prototype network can be anchored to the lesion without new annotation by lesion attraction on the prototype attention and a projection restricted to lesion cells without detected artifacts. In our same-backbone, leakage-controlled comparison with lesion-grouped statistics, the main lesion-anchored model reached an ISIC AUC not significantly different from that of the strongest mask-supervised ProtoPNet, kept all prototypes on the lesion, which no baseline did, and, under artifacts inserted around an unchanged lesion, flipped significantly fewer decisions than every baseline for dark corners and ink, never significantly more for any insertion, and kept the most stable explanation maps; an artifact-suppression term added nothing significant. The original mask loss generalized better externally, so combining it with lesion anchoring is a natural next step; the counterfactual protocol can check other interpretable classifiers as well.

References

[1] A. Esteva *et al.*, "Dermatologist-level classification of skin cancer with deep neural networks," *Nature*, vol. 542, no. 7639, pp. 115–118, 2017.

[2] R. Geirhos *et al.*, "Shortcut learning in deep neural networks," *Nat. Mach. Intell.*, vol. 2, pp. 665–673, 2020.

[3] J. K. Winkler *et al.*, "Association between surgical skin markings in dermoscopic images and diagnostic performance of a deep learning convolutional neural network for melanoma recognition," *JAMA Dermatol.*, vol. 155, no. 10, pp. 1135–1141, 2019.

[4] A. Bissoto *et al.*, "(De)constructing bias on skin lesion datasets," in *Proc. IEEE/CVF CVPR Workshops*, 2019.

[5] A. Bissoto, E. Valle, and S. Avila, "Debiasing skin lesion datasets and models? Not so fast," in *Proc. IEEE/CVF CVPR Workshops*, 2020.

[6] A. Bissoto *et al.*, "Artifact-based domain generalization of skin lesion models," in *Proc. ECCV Workshops*, 2022.

[7] S. Yan *et al.*, "EPVT: Environment-aware prompt vision transformer for domain generalization in skin lesion recognition," in *Proc. MICCAI*, 2023.

[8] S. W. Pewton and M. H. Yap, "Dark corner on skin lesion image dataset: Does it matter?" in *Proc. IEEE/CVF CVPR Workshops*, 2022, pp. 4831–4839.

[9] C. Rudin, "Stop explaining black box machine learning models for high stakes decisions and use interpretable models instead," *Nat. Mach. Intell.*, vol. 1, pp. 206–215, 2019.

[10] C. Chen *et al.*, "This looks like that: Deep learning for interpretable image recognition," in *Proc. NeurIPS*, vol. 32, 2019.

[11] A. Hoffmann *et al.*, "This looks like that... does it? Shortcomings of latent space prototype interpretability in deep networks," in *Proc. ICML Workshop on XAI*, 2021.

[12] M. Correia *et al.*, "XAI for skin cancer detection with prototypes and non-expert supervision," in *Proc. MICCAI Workshop iMIMIC*, 2023, arXiv:2402.01410.

[13] M. Nauta *et al.*, "Interpreting and correcting medical image classification with PIP-Net," in *Proc. ECAI Workshops*, CCIS vol. 1947, 2024, pp. 198–215.

[14] P. Bevan and A. Atapour-Abarghouei, "Skin deep unlearning: Artefact and instrument debiasing in the context of melanoma classification," in *Proc. ICML*, 2022.

[15] F. Pahde *et al.*, "Reveal to revise: An explainable AI life cycle for iterative bias correction of deep models," in *Proc. MICCAI*, 2023.

[16] A. J. Barnett *et al.*, "A case-based interpretable deep learning model for classification of mass lesions in digital mammography," *Nat. Mach. Intell.*, vol. 3, pp. 1061–1070, 2021.

[17] A. Bontempelli *et al.*, "Concept-level debugging of part-prototype networks," in *Proc. ICLR*, 2023.

[18] M. Nauta *et al.*, "Uncovering and correcting shortcut learning in machine learning models for skin cancer diagnosis," *Diagnostics*, vol. 12, no. 1, art. 40, 2022.

[19] X. Lin and M. Wright, "Diverse by design: Architectural constraints for prototype-based interpretability," arXiv:2609.27194, 2026.

[20] J. Adebayo *et al.*, "Sanity checks for saliency maps," in *Proc. NeurIPS*, vol. 31, 2018.

[21] A. S. Ross, M. C. Hughes, and F. Doshi-Velez, "Right for the right reasons: Training differentiable models by constraining their explanations," in *Proc. IJCAI*, 2017, pp. 2662–2670.

[22] L. Rieger *et al.*, "Interpretations are useful: Penalizing explanations to align neural networks with prior knowledge," in *Proc. ICML*, 2020, pp. 8116–8126.

[23] G. Huang *et al.*, "Densely connected convolutional networks," in *Proc. IEEE CVPR*, 2017, pp. 4700–4708.

[24] P. Tschandl, C. Rosendahl, and H. Kittler, "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions," *Sci. Data*, vol. 5, art. 180161, 2018.

[25] M. Combalia *et al.*, "BCN20000: Dermoscopic lesions in the wild," arXiv:1908.02288, 2019.

[26] N. C. F. Codella *et al.*, "Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC)," in *Proc. IEEE ISBI*, 2018, pp. 168–172.

[27] L.-C. Chen *et al.*, "Rethinking atrous convolution for semantic image segmentation," arXiv:1706.05587, 2017.

[28] A. Howard *et al.*, "Searching for MobileNetV3," in *Proc. IEEE/CVF ICCV*, 2019, pp. 1314–1324.

[29] T. Mendonça *et al.*, "PH^2 – A dermoscopic image database for research and benchmarking," in *Proc. IEEE EMBC*, 2013, pp. 5437–5440.

[30] T. Lee *et al.*, "DullRazor: A software approach to hair removal from images," *Comput. Biol. Med.*, vol. 27, no. 6, pp. 533–543, 1997.

[31] J. Kawahara *et al.*, "Seven-point checklist and skin lesion classification using multitask multimodal neural nets," *IEEE J. Biomed. Health Inform.*, vol. 23, no. 2, pp. 538–546, 2019.

[32] A. G. C. Pacheco *et al.*, "PAD-UFES-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones," *Data in Brief*, vol. 32, art. 106221, 2020.

[33] R. R. Selvaraju *et al.*, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," in *Proc. IEEE ICCV*, 2017, pp. 618–626.

[34] B. Zhou *et al.*, "Learning deep features for discriminative localization," in *Proc. IEEE CVPR*, 2016, pp. 2921–2929.

[35] A. Telea, "An image inpainting technique based on the fast marching method," *J. Graphics Tools*, vol. 9, no. 1, pp. 23–34, 2004.

[36] E. R. DeLong, D. M. DeLong, and D. L. Clarke-Pearson, "Comparing the areas under two or more correlated receiver operating characteristic curves: A nonparametric approach," *Biometrics*, vol. 44, no. 3, pp. 837–845, 1988.

[37] X. Sun and W. Xu, "Fast implementation of DeLong's algorithm for comparing the areas under correlated receiver operating characteristic curves," *IEEE Signal Process. Lett.*, vol. 21, no. 11, pp. 1389–1393, 2014.